

Detection & Analysis / Initial Triage

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SUMMARY:

Purpose: Quickly scope an incident while the system is still live. Collect volatile evidence before it disappears on reboot.

When: Right after an alert or suspicion of compromise.

Core Rule: Collect first. Analyze later.

Key Steps (in order):

1. **Documentation** — Start transcript and captains log (audit trail).
2. **System Info** — systeminfo or Get-ComputerInfo (OS, patches, uptime).
3. **Processes** — tasklist + **Process Explorer** (parent-child, suspicious parents, injection).
4. **Network** — netstat -ano (connections + PID), ipconfig /displaydns, arp -a.
5. **Users & Accounts** — net user, psloggedon, logonsessions.
6. **Persistence** — **Autoruns**, schtasks /query, services.
7. **Logs & Traffic** — Copy EVTX logs, capture with **Wireshark** or **RawCap**.

Main Tools: Process Explorer, Autoruns, netstat, Wireshark. **Why it matters:** Volatile data (processes, connections, memory) is lost on reboot. This phase decides if deeper host/network analysis is needed.

1. Detection & Analysis / Initial Triage

When to perform: Right after receiving an alert or suspecting compromise, while the system is still live (before reboot or shutdown). **Why:** Volatile data (processes, network connections, memory) disappears on reboot. This phase quickly identifies signs of compromise, captures evidence, determines scope, and decides if deeper analysis is needed. **Core rule:** **Collect evidence first. Analyze later.** Save all artifacts to a dedicated folder (\$datapath).

Investigation Workflow (Step-by-Step Order)

1. Start documentation (audit trail).
2. Gather basic system information (context).
3. Investigate running processes (malware runs as processes).
4. Check network connections and related artifacts (C2 communication).
5. Examine users, accounts, and logged-in sessions (unauthorized access).
6. Check services and persistence mechanisms (survival after reboot).
7. Collect logs and additional artifacts.
8. Capture network traffic if suspicious connections are found.

1. Documentation (PowerShell)

- **Tool:** PowerShell
PowerShell
Start-Transcript -Path \$datapath\transcript.txt
Why: Records every command you run and its output. Creates a defensible forensic audit trail of your actions.
- **Tool:** PowerShell
PowerShell
echo "captains_log: starting triage \$(Get-Date)" >> \$datapath\captains_log.txt
Why: Adds timestamped investigator notes for important observations.

2. Basic System Information (Built-in Windows / PowerShell)

- **Tool:** Command Prompt or PowerShell
cmd
systeminfo
or
PowerShell
Get-ComputerInfo
Why: Shows OS version, patch level, hostname, boot time, hardware, and domain membership. Reveals outdated systems, long uptime (possible compromise), or unusual configurations.

3. Running Processes Investigation

Processes are the most critical volatile artifact because malware must execute as a process.

- **Tool:** Built-in Windows (tasklist)
cmd
tasklist
Why: Lists processes, PIDs, and memory usage. Quickly spot strange names, high CPU/memory usage, or unknown executables.
- **Tool:** Sysinternals **Process Explorer** (procexp.exe) Run procexp.exe (GUI tool). **Why:** Displays parent-child relationships, loaded DLLs, network connections per process, and digital signatures. Detects process injection, suspicious parents (e.g., malware spawned from explorer.exe), or unsigned binaries.

4. Network Connection Investigation

Attackers need to communicate with C2 servers or other compromised systems.

- **Tool:** Built-in Windows (netstat)
cmd
netstat -ano
Why: Shows active connections, remote IPs/ports, states, and owning PID. Identify outbound C2 traffic or unusual internal connections.
Save output:
cmd
netstat -ano > \$datapath\netconn.txt
- **Tool:** Built-in Windows (ipconfig)
cmd
ipconfig /displaydns
Why: Displays recent DNS cache. Reveals malware domains, C2 servers, or suspicious lookups.
- **Tool:** Built-in Windows (arp)
cmd
arp -a
Why: Shows local devices recently communicated with. Helps detect lateral movement on the LAN.
- **Tool:** PowerShell
PowerShell
Get-NetRoute
Why: Shows routing table. Malware may manipulate routes to redirect traffic.
- **Tool:** PowerShell
PowerShell
Get-NetFirewallRule
Why: Lists firewall rules. Attackers often add rules to allow their traffic or bypass security.

5. User & Account Investigation

Attackers create or use accounts for persistence and privilege escalation.

- **Tool:** Built-in Windows (net)
cmd
net user
and
cmd
net localgroup administrators
Why: Lists accounts and admin group members. Spot new or suspicious users.
- **Tool:** Sysinternals (psloggedon.exe) Run psloggedon.exe. **Why:** Shows currently logged-on users (local and remote).
- **Tool:** Sysinternals (logonsessions64.exe) Run logonsessions64.exe. **Why:** Details session IDs, login times, and authentication types. Useful for correlating with event logs later.
- **Tool:** Built-in Windows (net)
cmd
net use
Why: Shows mapped network shares/drives. Ransomware often spreads via shares.

6. Services & Persistence Investigation

Malware installs as services or scheduled tasks to survive reboots.

- **Tool:** Built-in Windows (sc)
cmd
sc query
or PowerShell:
PowerShell
Get-WmiObject Win32_Service
Why: Lists services. Look for suspicious names or unknown binaries.
- **Tool:** Sysinternals **Autoruns** (autoruns.exe) Run autoruns.exe (GUI). **Why:** Scans startup items, scheduled tasks, services, drivers, and registry run keys. Finds persistence mechanisms.
- **Tool:** Built-in Windows (schtasks)
cmd
schtasks /query /fo LIST /v
Why: Lists scheduled tasks. Attackers use them to run malware or reconnect to C2.
- **Tool:** Sysinternals (listdlls.exe) Run listdlls.exe. **Why:** Shows loaded DLLs per process. Detects DLL injection or malicious libraries.
- **Tool:** Sysinternals (ntfsinfo.exe)
cmd
ntfsinfo C:
Why: Provides NTFS filesystem details. Useful for later timeline creation.

7. Log Collection & Traffic Capture

- **Tool:** PowerShell (copy)
PowerShell
Copy-Item C:\Windows\System32\winevt\Logs* \$datapath\winevent_logs
Why: Preserves raw EVTX logs (logins, process creation, etc.) for offline analysis.
- **Tool:** PowerShell (copy)
PowerShell
Copy-Item C:\Windows\System32\LogFiles* \$datapath\LogFiles
Why: Captures additional logs (firewall, IIS, etc.).
- **Tool:** **Wireshark** (GUI) Use filters such as: tcp.port == 8080 or ip.addr == <suspect IP> **Why:** Inspects live or captured traffic. Detects web shells or specific attacker commands.
- **Tool:** **RawCap** (rawcap.exe) — portable alternative when Wireshark cannot be installed. Run rawcap.exe to list interfaces, then capture. **Why:** Allows quick packet capture without installation.

Goal of phase: Quickly scope the incident, collect volatile evidence, and decide next steps (deeper host or network analysis).